Ajay Venkatesh

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Education

New York University, New York, NY — *Sep 2024 – May 2026*

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Experience

- **CampusMesh** — *Jan 2026 – Present*
  Software Engineer — New York, US
  - Developed scalable **Node.js microservices** powering real-time academic workflows across study groups, messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
  - Deployed **serverless architecture** on AWS using **Serverless Framework** with **Lambda, API Gateway, S3, and CloudFront** across **4 environments** (local, dev, staging, production); enforced security with **KMS-managed keys**, **SSM Parameter Store**, and **least-privilege IAM**.
  - Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.
- **Amazon** — *May 2025 – Aug 2025*
  Software Development Engineer Intern — Seattle, US
  - Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the **system design** for approval and established Role Based Access **(4+ roles)** using **OAuth, Midway SSO, and HELIS**.
  - Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by **70%** and enabling **multi-stage, multi-region CI/CD deployments**.
  - Built a **React 18** frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by **40%**.
  - Engineered backend services in **Java** using **Coral, Smithy, Dagger**, and **CBOR serialization**; applied low-level design and design patterns, reduced backend execution time by **35%** and ensured reliability by unit and integration testing using **Mockito**.
- **New York University (Novel AI Technologies, Inc.)** — *Aug 2024 – Present*
  Software Developer (On-Campus) — New York, US
  - Architected an **NSF-funded ($100K)** AI-powered health app for **iOS** and **Android**, processing real-time wearable data; deployed cloud-hosted AI models for health metrics analysis and food image processing.
  - Engineered ETL pipelines transforming **NoSQL** into **PostgreSQL** using **concurrency control and locking mechanisms**; processed **250K+** records and powered **AWS QuickSight** dashboards for real-time health analytics.
  - Developed an insurance analytics platform **(React, Node.js, REST APIs)**; deployed **Docker** containers on **EC2**, implemented Stripe billing, RBAC, and row-level security, generating **$50K+** revenue.
  - Streamlined underwriting via broker portal integrated with insurer platform; implemented **document and media processing** and **WebSocket-based real-time communication**; secured **6 insurance clients**.
- **PricewaterhouseCoopers (PwC)** — *Aug 2021 – Aug 2024*
  Senior Software Engineer — Bengaluru, India
  - Developed a distributed backend on **Microsoft Azure** for a health platform, integrating with **Microsoft CRM** and implementing event-driven processing and fault tolerance.
  - Led code reviews and mentored engineers on **software design principles**, reducing production bugs by **25%**.
  - Deployed an **NLP based OCR neural model** on Azure to automate structured data extraction, reducing data processing time by **75%**.
  - Optimized data ingestion with **parallel processing** for **100K**-row from Excel; reduced execution time from **5 hours to 1 hour**.
  - Built **Power BI dashboards** processing large datasets from multiple databases, improving reporting efficiency by **40%**.

Projects

- **Advanced Expense Tracker (Java Desktop Application)** | *Java, JavaFX, JDBC, SQLite, JFreeChart*
  - Architected a modular **Java application** supporting individual and group expenses with secure login, applying **OOP principles** and **MVC & DAO design patterns**.
  - Implemented concurrent transaction handling and real-time cost splitting using **multithreading** and synchronized blocks to ensure thread-safe database updates.
- **Hospital Management System (Full-Stack Application)** | *React, Node.js, Express, MongoDB, SageMaker, Hugging Face*
  - Built a **full-stack healthcare platform** using **React, Node.js, Express, and MongoDB** to manage appointments, patient records, lab reports, and medical history with **role-based access control**.
  - Fine-tuned a **Hugging Face transformer-based model** achieving **88% accuracy** for health condition prediction and deployed it on **AWS SageMaker** for real-time diagnostic support in the doctor dashboard.
- **RoBERTa Fine-Tuning with Custom LoRA** | *Python, PyTorch, Hugging Face, AGNews*
  - Architected a **parameter-efficient fine-tuning (PEFT)** pipeline using a custom **Low-Rank Adaptation (LoRA)** implementation to optimize a **RoBERTa-base** model for news classification.
  - Optimized model efficiency by freezing pretrained weights and injecting trainable rank-decomposition matrices, reducing trainable parameters to under **1M (934K)** while achieving **83.45%** accuracy on the private leaderboard.
  - Conducted extensive hyperparameter sweeps for rank (r) and alpha (α) values, utilizing **AdamW** and linear decay schedulers to stabilize training under strict memory and parameter constraints.

Skills

- **Programming Languages:** C++, Java, Python, TypeScript, JavaScript (ES6), SQL, C#
- **Web Technologies:** HTML, CSS, React, REST API, GraphQL API, Microservices, Node.js, Express.js, Sequelize
- **Database:** RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse
- **Cloud:** AWS (AWS CDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (OpenAI services, Function Apps, Security Data lakes, Storage blobs)
- **Tools:** Git, Visual Studio, Cursor, Jupyter, Postman, Docker, Kubernetes, Power BI, QuickSight, Jenkins, Ansible
- **AI/ML:** LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, RL

Certifications

AWS Cloud Practitioner | Azure Data Fundamentals | Stanford Machine Learning (Coursera)